Nonlocal and Deep Priors for Hyperspectral Anomaly Detection

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***Abstract*—Hyperspectral anomaly detection (HAD) seeks to identify targets of interest within hyperspectral images (HSIs) without requiring prior knowledge. Recent works show that embedding low-rank attributes in the HAD task can yield superior results. While most such methods focus on exploiting global or local geometrical structures in HSIs, largely overlooking similar features in nonlocal regions. To this end, we propose a novel nonlocal framework that captures the nonlocal self-similar (NSS) and global correlation along spectrum (GCS) priors within the subspace of HSIs. Then, we integrate a deep denoiser prior into the nonlocal framework to further leverage the external priors. The proposed approach, referred to as nonlocal and deep priors for anomaly detection (NLDPAD), combines the structural modeling of nonlocal low-rank and the feature extraction capabilities of deep learning (DL). Furthermore, an efficient alternating minimization (AM) algorithm was developed to optimize the framework, while the plug-and-play alternating direction method of multipliers (PnP-ADMM) was used to solve the subproblems. Extensive experiments on real satellite and aerial hyperspectral datasets demonstrate that the proposed NLDPAD framework significantly outperforms state-of-the-art methods, achieving superior detection performance.**

***Index Terms*—Alternating direction method of multipliers (ADMM), alternating minimization (AM), anomaly detection, deep models, hyperspectral image (HSI), nonlocal self-similarity (NSS), plug-and-play (PnP).**

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NOMENCLATURE

Abbreviation	Expansion
$\mathcal{Y}, \mathbf{Y}, \mathbf{y}$	Tensor, matrix, vector.
$\mathcal{R}_i\mathcal{Y} \in \mathbb{R}^{\sqrt{b}\times\sqrt{b}\times L\times k}$	Fourth-order tensor.
β, γ	Regularization parameters.
$\|\cdot\|_F$	Frobenius norm.
$\|\cdot\|_{2,1}$	$l_{2,1}$ norm.
$\|\cdot\|_{\mathbf{w}_i,*}$	Weighted nuclear norm.
$\mathcal{R}_i^T\mathcal{R}_i$	Number of overlapping patches that cover a specific pixel location.
$\mathcal{R}_i^T\mathbf{L}_i$	Sum of the values from all overlapping reconstructed patches at that pixel location.

I. INTRODUCTION

HYPERSPECTRAL imaging technology has revolutionized remote sensing by capturing detailed spectral information across a broad range of wavelengths for each pixel in an image [1], [2]. By acquiring hundreds of contiguous spectral bands for each pixel, hyperspectral images (HSIs) offer rich spectral information that facilitates the analysis and identification of materials, making it possible to detect subtle differences in surface properties [3], [4].

One of the key applications of this technology is hyperspectral anomaly detection (HAD), which focuses on identifying anomalous targets that differ from the background distribution within an HSI [1]. Unlike classification tasks [5], [6] that rely on predefined categories, HAD is an unsupervised approach and capable of identifying unforeseen or unknown targets without prior information. This is particularly valuable in high-risk or dynamic environments, such as military surveillance, precision agriculture, environmental monitoring, and disaster response, where unexpected materials or events may emerge without warning. To achieve effective separation of anomalies from the background, a variety of HAD methods have been developed, aiming to model the background and classify pixels that deviate from this model as anomalies. Current approaches in HAD are primarily grouped into statistical-based, deep learning (DL)-based, and reconstruction-based techniques.

The statistical-based methods assume that the background follows a specific probability distribution, identifying pixels that deviate from this distribution as anomalies [7]. A benchmark example of this approach is the RX detector [8], which

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models the background using a multivariate Gaussian distribution and detects anomalies in HSIs using the Mahalanobis distance metric. Numerous RX variants, such as the weighted RX [9], enhance background modeling by assigning weights to different image components. The kernel RX method [10] applies kernel techniques to improve discrimination between background and anomalies. More recently, Liu et al. [11] proposed the 2S-GLRT method, which uses a dual-window approach to detect multipixel anomalies in HSIs. While effective in straightforward scenarios, these methods often face challenges in complex real-world environments due to their reliance on predefined statistical models.

The DL methods have found broad applications in various remote sensing fields, and HAD is no exception. DL-based detectors leverage neural networks to extract deep features from HSIs, using architectures such as autoencoders (AEs) [12], [13], generative adversarial networks (GANs) [14], and convolutional neural networks (CNNs) [15]. Most AE and GAN-based HAD methods operate in an unsupervised learning paradigm, where all pixels or patches of HSIs are input to the network for reconstruction, with the goal of accurately modeling the background. Reconstruction errors are then interpreted as indicators of anomalous probabilities. For instance, Fan et al. [16] added a graph regularization constraint to the hidden layer of an AE, creating the RGAE model to enhance network robustness for background modeling and capture spatial features in HSIs. Xiang et al. [17] proposed a guided model to improve background reconstruction. Recently, blind-spot network models based on CNN frameworks have been applied to HAD, enabling cleaner background reconstruction in a self-supervised manner [18], [19]. Wang et al. [20] introduced the blind-block reconstruction network (BockNet), which uses a guard window in the receptive field center to prevent interference from anomalous targets, enabling effective detection of anomalies of various sizes. An unsupervised framework based on the multivariate probability distribution autoencoder (MPDA) [21] has been developed to learn the intrinsic features of HSI without requiring explicit labels. A novel nonlocal and local feature-coupled self-supervised network (NL2Net) [22] has been developed, which adopts a blind-spot network framework to effectively couple nonlocal and local features of data under a self-supervised learning paradigm. A multiscale frequency-guided two-stream network (MFTNet) [23] has been proposed, which recovers the background use a low-frequency recovery branch and enhance the abnormal features through a high-frequency enhancement branch. Despite their effectiveness, DL methods for HAD involve a large number of training parameters and are often associated with high computational costs.

With advances in compressed sensing and machine learning [24], [25], the reconstruction-based methods for HAD, including sparse representation (SR), collaborative representation (CR), and low-rank representation (LRR), have garnered increasing interest as they do not rely on background distribution assumptions. The SR-based detector [26] assumes that normal pixels in an image can be represented by a limited number of atoms from an overcomplete dictionary, whereas anomalies cannot. In contrast, the CR-based method [27] assumes that background pixels can be represented by their spatial surroundings, achieved through a linear combination of neighboring pixels around each test pixel. Both the SR and CR approaches perform anomaly detection based on the reconstruction error between the recovered background and the original HSIs. Generally, both the approaches perform pixel-by-pixel detection, overlooking the global structure inherent in HSIs.

By leveraging the global structure information of HSIs, the LRR-based approaches have been widely applied in HAD. One example is the robust principal component analysis (RPCA) model [28], which assumes background pixels arise from a single subspace. Sun et al. [29] introduced the RPCA-RX method, decomposing the HSIs into low-rank and sparse components, followed by using the RX detector on the sparse part to derive anomalies. Zhang et al. [30] developed the low-rank and sparse-decomposition-based Mahalanobis distance (LSMAD) method, which uses the Mahalanobis distance for similarity measurement. Xu et al. [31] designed the low-rank and SR (LRASR) detection method with sparse constraints introduced to characterize the local structure. Meanwhile, joint K-means and RX techniques are performed to construct typical background dictionaries. To further acquire stable results, Niu and Wang [32] suggested a learnable dictionary construction method. Following LRASR, LSDM-MoG [33] converts the sparse part into the form of a mixture of Gaussian models to tackle the complex noise in HSIs. Jiang et al. proposed a hierarchical metric-learning-based method, which can optimize intra- and interclass relationships to enhance the accuracy [34]. To fully integrate spatial information, tensor decomposition and total variation (TV) regularization have been incorporated into the LRR model. Cheng and Wang [35] embed the graph and TV regularization into LRR aiming at capturing the geometrical structure information of HSIs. The tensor principal component analysis (TPCA)-based approach can effectively prevent background interference [36], [37]. Wang et al. [38] later developed a tensor LRR-based HAD method to directly decompose HSIs into background and anomaly tensors. A unified nonconvex HAD framework using the generalized shrinkage mappings (HADGSMs) has been proposed in [39] to provide a more precise approximation of the LRR model, enabling a more accurate distinction between background and anomaly components. Recent work by Fu et al. [40] proposed a plug-and-play (PnP) denoising CNN regularized anomaly detection (DeCNN-AD) method, which combines the conventional LRR and CNN prior to further improving detection performance. Xue et al. [41] proposed the multiple-layer representation with Tucker decomposition and CANDECOMP/PARAFAC (CP) decomposition [42] to better explore the hierarchical attributes hidden in high-dimensional images. Later in [43], a tensor convolution-like low-rank dictionary paradigm has been developed to further explore the shift-invariant low-dimensional structure of high-dimensional images.

While low-rank modeling can globally characterize strong correlations in images, they have largely ignored nonlocal self-similar (NSS) properties presented in images. Recent years have witnessed increasing attention in nonlocal

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low-rank modeling for image reconstruction [44], [45], [46], which involves grouping similar or highly correlated patches from different locations within an image. Zhuang et al. [47] introduced a global and nonlocal low-rank factorization method for HSI denoising by exploiting the high correlation and self-similarity inherent within HSI. Subsequently, in [48], a fast and parameter-free method called FastHyMix was proposed for removing mixed noise in HSIs. FastHyMix models the complex noise distribution using a Gaussian mixture model, while simultaneously exploiting two key characteristics of HSI: low-rank structure in the spectral domain and high spatial correlation. More recent nonlocal meets global (NGmeet) method [49] exploits the spectral and spatial properties of HSIs, demonstrating promising performance in various HSI restoration tasks, such as denoising, inpainting, and compressed HSI reconstruction. Motivated by the outstanding performance of nonlocal low-rank modeling, this article develops new nonlocal and deep priors for anomaly detection (NLDPAD) framework, which integrates the traditional nonlocal low-rank modeling framework with a deep prior. A low-rank modeling of nonlocal similar 3-D patches is developed to capture the NSS and global correlation along spectrum (GCS) priors in the subspace of the HSIs. Then, a deep denoiser prior is plugged into the nonlocal anomaly detection (NLAD) framework to further incorporate the external priors. Compared with the existing LRR-based techniques, the proposed NLDPAD approach can enhance detection performance by leveraging both the structural modeling of LRR and the feature extraction capabilities of DL. We should mention that a key aspect of the newly proposed NLDPAD framework is that with its open structure, it is able to accept different types of nonlocal or deep priors, thus providing significant flexibility for exploring various types of information within HSIs. The main contributions of this work are summarized as follows.

1) A new NLDPAD paradigm that can integrate both nonlocal low-rank modeling with deep priors, which is the first time developed and tailored for HAD tasks. The proposed paradigm belongs to the PnP framework, allowing for enhanced anomaly detection in HSIs by jointly capturing NSS, GCS, and additional external priors.
2) To the best of our knowledge, this is the first work to extend the nonlocal low-rank modeling in the subspace of the HSIs for the HAD task.
3) An efficient alternating minimization (AM) algorithm was developed to optimize the proposed framework. The PnP alternating direction method of multipliers (PnP-ADMM) was designed to optimize the subproblem.
4) We conduct extensive experiments on both the real satellite and aerial hyperspectral datasets to validate the superior detection performance of the proposed approach.

The remainder of this work is organized as follows. Section II reviews the related work. Section III details the proposed framework and solving algorithm. Section IV presents experimental results on both the real satellite and aerial hyperspectral datasets. The conclusion is drawn in Section V.

II. Notations and Related Work

In this section, we first give the basic notations, as presented in Nomenclature, and then give a brief introduction to the reconstruction-based model, PnP model, and nonlocal low-rank tensor recovery model.

A. Reconstruction-Based Model

The goal of the reconstruction-based model for HAD is to split the original HSIs into the background part and anomaly part, which can be modeled as [1]

$$\mathbf{Y} = \mathbf{DX} + \mathbf{E} \tag{1}$$

where $\mathbf{Y} := [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_n] \in \mathbb{R}^{L\times n}$ represents the observed HSIs with n pixels and each $\mathbf{y}_i \in \mathbb{R}^{L\times 1}$ represents the L-spectrum measured pixel; $\mathbf{DX}$ represents the background component, $\mathbf{D} := [\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_m] \in \mathbb{R}^{L\times m}$ and $\mathbf{X} := [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n] \in \mathbb{R}^{m\times n}$ represent the dictionary and corresponding representation coefficients, where the dictionary $\mathbf{D}$ is chosen from the background pixels; and $\mathbf{E} := [\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n] \in \mathbb{R}^{L\times n}$ represents the anomaly component.

To accurately separate the anomalies and background from the original HSIs, the prior information of $\mathbf{X}$ and $\mathbf{E}$ are usually incorporated into the following framework:

$$\begin{aligned} &\min_{\mathbf{X},\mathbf{E}} \ \Xi(\mathbf{X}) + \gamma\, \Psi(\mathbf{E}) \\ &\text{s.t. } \mathbf{Y} = \mathbf{DX} + \mathbf{E} \end{aligned} \tag{2}$$

where $\Xi(\cdot)$ and $\Psi(\cdot)$ represent the regularization terms of the representation coefficients and anomaly component, respectively. Many regularizers, such as low-rank [31], [50], graph [35], TV [51], [52], and generalized shrinkage mappings [53], are investigated to exploit the global or local geometrical structure and spatial relationships in HSIs.

B. PnP Model

Recent studies have shown that the PnP model offers a highly adaptable framework for addressing various image reconstruction challenges such as image deblurring [54], image compressive sensing [46], and image inpainting [55], by incorporating an off-the-shelf image denoiser. The core of the PnP framework aims at decoupling the problem-specific sensing modality from general image priors, making it a flexible approach for tackling complex image reconstruction problems, which can be expressed as

$$\min_{\mathbf{X}} \ \frac{1}{2}\|\mathbf{Y} - f(\mathbf{X})\|_F^2 + \beta\, \Theta(\mathbf{X}) \tag{3}$$

where f represents the degradation operator. The first term in the model (3) represents the fidelity term, while the second term $\Theta(\cdot)$ serves as the regularizer based on the chosen image denoiser priors. Many off-the-shelf image denoisers, such as K-SVD [56], BM3D [57], and weighted nuclear norm (WNN) [58], can be adapted. As a result, unlike traditional model-based methods that require explicitly defined or handcrafted image priors, the PnP framework implicitly defines priors through the denoiser. Such an advantage allows for the integration of advanced deep denoisers, thus enhancing overall effectiveness.

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C. Nonlocal Low-Rank Tensor Recovery Model

The NSS framework has been extensively explored in many image restoration applications [45], aiming to reconstruct the original image from its degraded observation. The NSS prior indicates that the HSI patches usually resemble other nonlocal structures within the same image. The NSS-based methods typically group similar patches and use the correlations within each group to enhance image restoration. Specifically, a set of overlapping full-band patches $\Omega := \{\mathcal{Y}_i\}_{i=1}^{b} \in \mathbb{R}^{\sqrt{b}\times\sqrt{b}\times L}$ are extracted from the tensor formed HSIs $\mathcal{Y} \in \mathbb{R}^{n_r\times n_r\times L}(n = n_r \times n_c)$. Then a set of patches $\mathcal{R}_i\mathcal{Y} := \{\mathcal{Y}_{i,1}, \mathcal{Y}_{i,2}, \ldots, \mathcal{Y}_{i,k}\}$ that are most similar to each example patch $\mathcal{Y}_i$ can be obtained by performing block matching [59], where each operator $\mathcal{R}_i : \mathcal{Y} \to \mathcal{R}_i\mathcal{Y} \in \mathbb{R}^{\sqrt{b}\times\sqrt{b}\times L\times k}$ applies a K-nearest neighbor (KNN) [60] algorithm to select the k closest patches based on Euclidean distance. By stacking these patches and reshaping each $\mathcal{R}_i\mathcal{Y}$ to form the nonlocal tensor group $\mathcal{R}_i\mathbf{Y}$, we can then recover the image through the following nonlocal low-rank tensor recovery model:

$$\min_{\mathbf{O}_i} \sum_i \frac{1}{\lambda_i^2}\|\mathcal{R}_i\mathbf{Y} - \mathbf{O}_i\|_F^2 + \Phi(\mathbf{O}_i) \tag{4}$$

where λ_i^2 represents the noise variance of the corresponding constructed image; $\mathbf{O}_i$ represents the corresponding original nonlocal similar patch, and $\Phi(\cdot)$ represents the low-rank regularizer. Common choices for $\Phi(\cdot)$ include rank penalty [61], nuclear norm [62], and WNN [63]. Note that both the NSS and GCS priors are preserved and reflected by the nonlocal low-rank tensor recovery model, along with its nonlocal similar patch number and spectral modes, respectively.

III. Proposed Approach and Solving Algorithm

In this section, we first give the model formulation of the proposed approach and then derive the solving algorithm.

A. Model Formulation

Inspired by the NSS framework which shows that using the low-rank attribute embedding to conduct the restoration tasks can remarkably restore the image, we further take advantage of this strategy to make it more applicable in anomaly detection tasks. Step by step, we first expound on the motivation for using the NSS framework to conduct the anomaly detection task in detail. As mentioned above, the goal of the reconstruction-based model for anomaly detection is to accurately separate the anomalies and background as much as possible. In other words, the better the background part is reconstructed, the more accurately anomalies can be isolated. While regularization techniques that exploit global or local geometrical structures have shown promise in reconstructing background parts, they often fail to leverage the recurring patterns and textures found in nonlocal regions, which are common in real-world scenes. The nonlocal low-rank tensor recovery model exploits the NSS and GCS priors via the original image but is not applicable for anomaly detection tasks. We start from the strategy that the HSIs should lie in a low-dimensional spectral subspace, which has been extensively accepted for hyperspectral denoising [49], compressed sensing [64], and hyperspectral restoration [65]. Motivated by this viewpoint, we expect to capture the NSS and GCS priors in the subspace of the HSIs. Therefore in this work, we develop an NLAD model, which can be formulated as follows:

$$\begin{aligned}&\min_{\mathbf{X},\mathbf{L}_i,\mathbf{E}} \sum_i \frac{1}{\lambda_i^2}\|\mathcal{R}_i\mathbf{X} - \mathbf{L}_i\|_F^2 + \Phi(\mathbf{L}_i) + \gamma\,\Psi(\mathbf{E})\\ &\text{s.t. } \mathbf{Y} = \mathbf{D}\mathbf{X} + \mathbf{E}\end{aligned} \tag{5}$$

where $\mathcal{R}_i\mathbf{X}$ represents each nonlocal similar patch group of the representation coefficients selected by the KNN algorithm, and $\mathbf{L}_i$ represents the corresponding nonlocal similar patch.

The NLAD model can only leverage the inherent internal features of HSIs, which may result in poor performance. To further incorporate other complementary prior, we developed a new NLDPAD paradigm by combining the nonlocal and deep denoiser priors, which jointly leverage the NSS, GCS, and external priors. Specifically, the NLDPAD model can be formulated as follows:

$$\begin{aligned}&\min_{\mathbf{X},\mathbf{L}_i,\mathbf{E}} \sum_i \frac{1}{\lambda_i^2}\|\mathcal{R}_i\mathbf{X} - \mathbf{L}_i\|_F^2 + \Phi(\mathbf{L}_i) + \beta\,\Theta(\mathbf{X}) + \gamma\,\Psi(\mathbf{E})\\ &\text{s.t. } \mathbf{Y} = \mathbf{D}\mathbf{X} + \mathbf{E}\end{aligned} \tag{6}$$

where the goal is to optimize the representation coefficients of the background dictionary $\mathbf{X}$, the corresponding nonlocal similar patch $\mathbf{L}_i$, and the anomaly component $\mathbf{E}$, while the observed HSIs $\mathbf{Y}$ and the background dictionary $\mathbf{D}$ are given. Similar to [31], [35], and [39], the proposed framework constructs the background dictionary in advance. In this approach, K-means clustering with Euclidean distance is first used to divide the HSIs into M clusters. RX anomaly detection is then applied to each cluster. For each test pixel, the square of the Mahalanobis distance to the local background mean within the cluster is calculated. The P pixels with the smallest Mahalanobis distances are selected as the dictionary atoms. The regularization terms of the proposed NLDPAD model are described below.

1) Nonlocal Similar Patch Regularization $\Phi(\mathbf{L}_i)$: Nonlocal similar patches exhibit strong structural correlations. In this work, we use the WNN as the low-rank regularizer, as the corresponding method [63] has demonstrated superior performance in image restoration compared with other approaches. The nonlocal similar patch regularizer is defined as

$$\Phi(\mathbf{L}_i) = \sum_i \|\mathbf{L}_i\|_{\mathbf{w}_i,*} \tag{7}$$

where

$$\|\mathbf{L}_i\|_{\mathbf{w}_i,*} = \sum_{j=1}^{d} w_{ij}\sigma_{ij} \tag{8}$$

with $\mathbf{w}_i = [w_{i1}, \ldots, w_{id}]^T$ and $d = \min(bL, k)$, $\forall j = 1, \ldots, d$. $w_{ij} \geq 0$ is the nonnegative weight assigned to σ_{ij} and σ_{ij} is the jth singular value of the corresponding nonlocal similar patch $\mathbf{L}_i$.

2) Image Denoiser Regularization $\Theta(\mathbf{X})$: Instead of designing the new regularizer to exploit the global geometrical structure of the representation coefficients, we directly use an existing regularizer from the off-the-shelf denoiser. In this

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work, the prior of a denoising network called fast and flexible denoising CNN (FFDNet) [66] is plugged in the model (6). We should mention that the FFDNet is chosen for this work due to several key advantages over other denoisers. Compared with machine-learning-based denoisers [56], [58], FFDNet, once fully trained, offers significantly faster performance. In addition, unlike other DL-based denoisers [67], [68], FFDNet can handle images with varying noise levels without requiring retraining, making it more versatile for different noise conditions. The details of the FFDNet architecture used in this study are as follows. The process begins with a reversible downsampling operator, which reshapes the representation coefficient $\mathbf{X}$ into four downsampled subimages. These subimages are then concatenated with a tunable noise level map to form the input to the network. The core of the model consists of a depth of 15 with a sequence of 3×3 convolution layers. Each layer uses a specific combination of the following operations: convolution (Conv), batch normalization (BN), and rectified linear units (ReLUs). Specifically, the first layer uses "Conv + ReLU," the intermediate layers use "Conv + BN + ReLU," and the final layer uses a single "Conv" operation. Zero-padding is applied to preserve the spatial dimensions of the feature maps after each convolution. Following the final convolutional layer, an upscaling operation is performed, acting as the inverse of the initial downsampling operator, to reconstruct the estimated representation coefficient.

3) Anomaly Component Regularization $\Psi(\mathbf{E})$: In real scenarios, only a small fraction of pixels belong to anomalies, making the matrix $\mathbf{E}$ sparse. We use the $l_{2,1}$-norm as the anomaly component regularizer, which is defined as

$$\Psi(\mathbf{E}) = \|\mathbf{E}\|_{2,1} \tag{9}$$

where

$$\|\mathbf{E}\|_{2,1} = \sum_{i=1}^{n} \sqrt{\sum_{j=1}^{L} \mathbf{E}_{ij}^2} \tag{10}$$

and $\mathbf{E}_{ij}$ represents the entry of $\mathbf{E}$. The $l_{2,1}$-norm promotes sparsity in the columns of $\mathbf{E}$, implying that the corruptions are "sample-specific"—some pixels are affected, while others remain unaffected. A larger magnitude in the column vector corresponding to the ith pixel indicates a higher degree of anomaly for that pixel. As a result, the matrix $\mathbf{E}$ inherently captures and quantifies the anomalies in the image.

Remark 1: The existing LRR-based anomaly detection approaches decompose the image into a low-rank background component and a sparse anomalous component, which can effectively capture the global spectral correlations in HSI. However, such methods overlook the NSS properties presented in HSI, thereby failing to exploit the recurring patterns and textures in nonlocal regions. In contrast, the proposed nonlocal modeling groups similar patches and uses the correlations within each group in the subspace of the HSI, enabling the simultaneous capture of both the NSS and GCS priors. The nonlocal modeling increases computational cost due to the need for searching similar patch groups. This reflects a fundamental tradeoff: while incorporating richer prior information enhances detection performance, it also demands greater computational resources.

B. AM Algorithm

It is quite arduous to solve the NLDPAD model directly due to multiple variables existing in the objective function. To address this, we develop an AM algorithm that alternately iteratively estimates $\mathbf{L}_i$ and $(\mathbf{X}, \mathbf{E})$, simplifying the optimization process. As will be demonstrated in the sequel, the optimization problem for $\mathbf{L}_i$ has the closed-form solution. For the optimization problem involving $(\mathbf{X}, \mathbf{E})$, we further develop a PnP-ADMM to solve it efficiently. The overall illustration of the proposed framework is presented in Fig. 1.

1) Optimization of $\mathbf{L}_i$: For fixed $(\mathbf{X}, \mathbf{E})$, the optimization problem (6) with respect to $\mathbf{L}_i$ can be reduced as follows:

$$\mathbf{L}_i^* = \underset{\mathbf{L}_i}{\operatorname{argmin}} \sum_i \frac{1}{\lambda_i^2} \|\mathcal{R}_i\mathbf{X} - \mathbf{L}_i\|_F^2 + \sum_i \|\mathbf{L}_i\|_{\mathbf{w}_i,*}. \tag{11}$$

Then for each group patch $\mathcal{R}_i\mathbf{X}$, the optimal $\mathbf{L}_i^*$ can be achieved by the well-known singular values thresholding [63]

$$\mathbf{L}_i^* = \mathbf{P}_i \mathcal{D}_{\mathbf{w}_i}(\Delta_i) \mathbf{Q}_i^T \quad \forall i \tag{12}$$

where $\mathcal{R}_i\mathbf{X} = \mathbf{P}_i\Delta_i\mathbf{Q}_i^T$ denotes the singular value decomposition of $\mathcal{R}_i\mathbf{X}$, and $\mathcal{D}_{\mathbf{w}_i}(\Delta_i)$ denotes the generalized soft-thresholding operator with the weighted vector $\mathbf{w}_i$, i.e., $\mathcal{D}_{\mathbf{w}_i}(\Delta)_{ij} = \max(\delta_{ij} - \lambda_i^2 w_{ij}, 0)$ with δ_{ij} being the jth diagonal element of Δ_i.

2) Optimization of $(\mathbf{X}, \mathbf{E})$: For fixed $\mathbf{L}_i$, the optimization problem (6) with respect to $(\mathbf{X}, \mathbf{E})$ can be reduced as follows:

$$\begin{aligned}(\mathbf{X}^*, \mathbf{E}^*) = \underset{\mathbf{X},\mathbf{E}}{\operatorname{argmin}} \sum_i \frac{1}{\lambda_i^2} \|\mathcal{R}_i\mathbf{X} - \mathbf{L}_i\|_F^2 + \beta\, \Theta(\mathbf{X}) + \gamma\, \|\mathbf{E}\|_{2,1} \\ \text{s.t. } \mathbf{Y} = \mathbf{D}\mathbf{X} + \mathbf{E}.\end{aligned} \tag{13}$$

Note that the variables $\mathbf{X}$ and $\mathbf{E}$ in model (13) are coupled; moreover, it also involves the deep prior term. The PnP-ADMM [69] uses a specific manner to split the deep prior variables, demonstrating promising results in various applications. Specifically, by introducing two slack variables $\mathbf{Z}_1 \in \mathbb{R}^{m\times n}$ and $\mathbf{Z}_2 \in \mathbb{R}^{m\times n}$, model (13) can be equivalently rewritten as

$$\begin{aligned}\underset{\mathbf{X},\mathbf{E},\mathbf{Z}_1,\mathbf{Z}_2}{\operatorname{argmin}} \sum_i \frac{1}{\lambda_i^2} \|\mathcal{R}_i\mathbf{Z}_1 - \mathbf{L}_i\|_F^2 + \beta\, \Theta(\mathbf{Z}_2) + \gamma\, \|\mathbf{E}\|_{2,1} \\ \text{s.t. } \mathbf{Y} = \mathbf{D}\mathbf{X} + \mathbf{E},\ \mathbf{Z}_1 = \mathbf{X},\ \mathbf{Z}_2 = \mathbf{X}.\end{aligned} \tag{14}$$

Let $\sigma > 0$ be a penalty parameter, the augmented Lagrangian function of model (14) is defined by

$$\begin{aligned}&L_\sigma(\mathbf{X}, \mathbf{E}, \mathbf{Z}_1, \mathbf{Z}_2; \mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3) \\ &= \sum_i \frac{1}{\lambda_i^2} \|\mathcal{R}_i\mathbf{Z}_1 - \mathbf{L}_i\|_F^2 \\ &\quad + \beta\, \Theta(\mathbf{Z}_2) + \gamma\, \|\mathbf{E}\|_{2,1} + \frac{\sigma}{2} \|\mathbf{Z}_1 - \mathbf{X} - \sigma^{-1}\mathbf{V}_1\|_F^2 + \frac{\sigma}{2} \|\mathbf{Z}_2 \\ &\quad - \mathbf{X} - \sigma^{-1}\mathbf{V}_2\|_F^2 + \frac{\sigma}{2} \|\mathbf{Y} - \mathbf{D}\mathbf{X} - \mathbf{E} - \sigma^{-1}\mathbf{V}_3\|_F^2 - \frac{1}{2\sigma} \|\mathbf{V}_1\|_F^2 \\ &\quad - \frac{1}{2\sigma} \|\mathbf{V}_2\|_F^2 - \frac{1}{2\sigma} \|\mathbf{V}_3\|_F^2\end{aligned} \tag{15}$$

where $\mathbf{V}_1 \in \mathbb{R}^{m\times n}$, $\mathbf{V}_2 \in \mathbb{R}^{m\times n}$, and $\mathbf{V}_3 \in \mathbb{R}^{L\times n}$ are Lagrangian dual variables. Now we can solve problem (15) under the PnP-ADMM framework. As will be shown in

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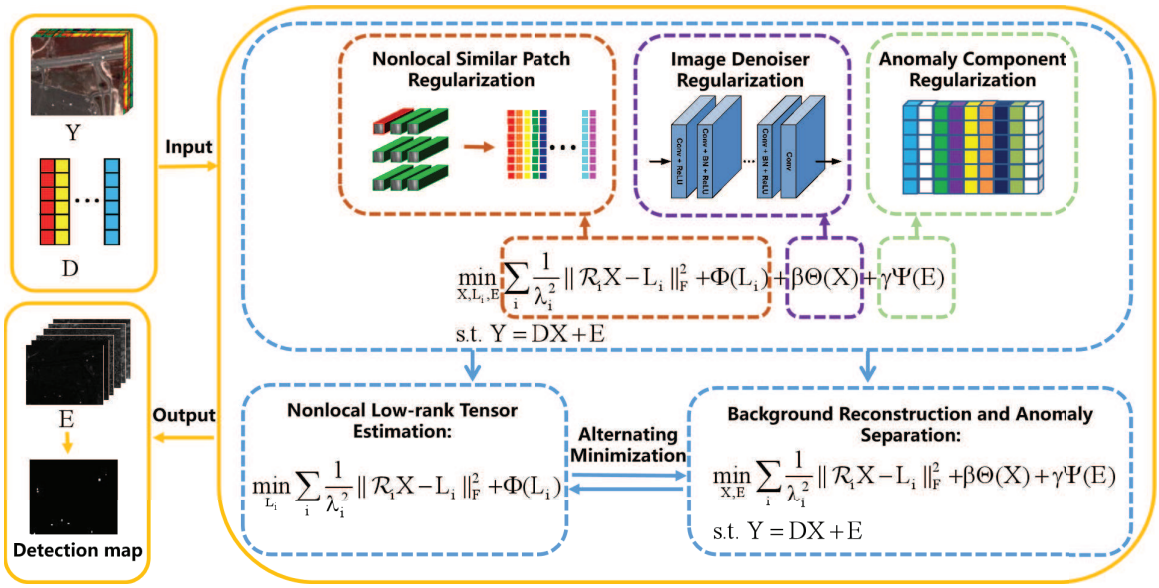


Fig. 1. Overall illustration of the proposed framework.

the sequel, the PnP-ADMM alternatively minimizes the augmented Lagrangian function (15) with respect to different variables, followed by the update of Lagrangian dual variables.

a) X subproblem: For fixed other variables, the $\mathbf{X}$ subproblem of model (15) can be written as

$$\mathbf{X}^* = \underset{\mathbf{X}}{\operatorname{argmin}} \frac{\sigma}{2}\|\mathbf{Z}_1 - \mathbf{X} - \sigma^{-1}\mathbf{V}_1\|_F^2 + \frac{\sigma}{2}\|\mathbf{Z}_2 - \mathbf{X} - \sigma^{-1}\mathbf{V}_2\|_F^2 + \frac{\sigma}{2}\|\mathbf{Y} - \mathbf{D}\mathbf{X} - \mathbf{E} - \sigma^{-1}\mathbf{V}_3\|_F^2. \quad (16)$$

It is clear that (16) is a quadratic optimization problem, and its solution can be expressed as

$$\mathbf{X}^* = \left(\mathbf{D}^T\mathbf{D} + 2\mathbf{I}\right)^{-1}\left(\mathbf{Z}_1 - \sigma^{-1}\mathbf{V}_1 + \mathbf{Z}_2 - \sigma^{-1}\mathbf{V}_2 - \mathbf{D}^T\mathbf{E} - \sigma^{-1}\mathbf{D}^T\mathbf{V}_3 + \mathbf{D}^T\mathbf{Y}\right) \quad (17)$$

where $\mathbf{I}$ denotes the identity matrix.

b) E subproblem: By fixing other variables, the $\mathbf{E}$ subproblem of model (15) is performed as

$$\mathbf{E}^* = \underset{\mathbf{E}}{\operatorname{argmin}}\, \gamma\, \|\mathbf{E}\|_{2,1} + \frac{\sigma}{2}\|\mathbf{Y} - \mathbf{D}\mathbf{X} - \mathbf{E} - \sigma^{-1}\mathbf{V}_3\|_F^2. \quad (18)$$

Then the solution of model (18) can be obtained by the well-known vector-soft threshold [70], [71] in a closed form as

$$\mathbf{E}^* = S_{\gamma/\sigma}\left(\mathbf{Y} - \mathbf{D}\mathbf{X} - \sigma^{-1}\mathbf{V}_3\right) \quad (19)$$

where

$$S_a(\mathbf{C})(:,i) := \left(\frac{\max\{\|\mathbf{C}(:,i)\|_2 - a, 0\}}{\max\{\|\mathbf{C}(:,i)\|_2 - a, 0\} + a}\right)\mathbf{C}(:,i). \quad (20)$$

Here, $S_a(\mathbf{C})(:,i)$ and $\mathbf{C}(:,i)$ denote the ith column of $S_a(\mathbf{C})$ and $\mathbf{C}$, respectively.

c) $\mathbf{Z}_1$ subproblem: The optimization problem with respect to $\mathbf{Z}_1$ can be performed as

$$\mathbf{Z}_1^* = \underset{\mathbf{Z}_1}{\operatorname{argmin}} \sum_i \frac{1}{\lambda_i^2}\|\mathcal{R}_i\mathbf{Z}_1 - \mathbf{L}_i\|_F^2 + \frac{\sigma}{2}\|\mathbf{Z}_1 - \mathbf{X} - \sigma^{-1}\mathbf{V}_1\|_F^2. \quad (21)$$

By simple calculation, $\mathbf{Z}_1^*$ can be solved with the following closed-form solution:

$$\mathbf{Z}_1^* = \left(\frac{2}{\lambda_i^2}\sum_i \mathcal{R}_i^T\mathcal{R}_i + \sigma\mathbf{I}\right)^{-1}\left(\frac{2}{\lambda_i^2}\sum_i \mathcal{R}_i^T\mathbf{L}_i + \sigma\mathbf{X} + \mathbf{V}_1\right) \quad (22)$$

where $\mathcal{R}_i^T\mathcal{R}_i$ represents the number of overlapping patches that cover a specific pixel location, and $\mathcal{R}_i^T\mathbf{L}_i$ represents the sum of the values from all overlapping reconstructed patches at that pixel location.

d) $\mathbf{Z}_2$ subproblem: Given $\mathbf{X}$, $\mathbf{E}$, and $\mathbf{Z}_1$, then the $\mathbf{Z}_2$ subproblem in model (15) can be rewritten as

$$\mathbf{Z}_2^* = \underset{\mathbf{Z}_2}{\operatorname{argmin}} \beta\,\Theta(\mathbf{Z}_2) + \frac{\sigma}{2}\|\mathbf{Z}_2 - \mathbf{X} - \sigma^{-1}\mathbf{V}_2\|_F^2. \quad (23)$$

Note that the optimization problem (23) is the proximity operator of $\Theta(\cdot)$ applied to $\hat{\mathbf{Z}} := \mathbf{X} + \sigma^{-1}\mathbf{V}_2$. Accordingly, we express this problem as

$$\mathbf{Z}_2^* = \mathcal{F}\left(\hat{\mathbf{Z}}, \beta/\sigma\right) \quad (24)$$

where $\mathcal{F}$ represents a Gaussian denoiser based on the deep prior term $\Theta(\cdot)$. In general, any advanced deep denoiser can be adopted as $\mathcal{F}$. In this work, we adopted the effective and efficient FFDNet as the deep prior for background reconstruction. In the FFDNet, parameter β is related to the input's standard deviation of Gaussian noise.

e) $\mathbf{V}_1$, $\mathbf{V}_2$, and $\mathbf{V}_3$ subproblems: During the iteration, the Lagrangian dual variables $\mathbf{V}_1$, $\mathbf{V}_2$, and $\mathbf{V}_3$ are updated as follows:

$$\begin{aligned}\mathbf{V}_1^* &= \mathbf{V}_1 - \tau\sigma(\mathbf{Z}_1 - \mathbf{X})\\ \mathbf{V}_2^* &= \mathbf{V}_2 - \tau\sigma(\mathbf{Z}_2 - \mathbf{X})\\ \mathbf{V}_3^* &= \mathbf{V}_3 - \tau\sigma(\mathbf{Y} - \mathbf{D}\mathbf{X} - \mathbf{E}).\end{aligned} \quad (25)$$

Once the optimization for (6) is complete, the anomaly detection task is carried out on the anomaly component. More

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Algorithm 1 Proposed AM Algorithm for NLDPAD Model

Input: HSI $\mathbf{Y}$, background dictionary $\mathbf{D}$, and the regularization parameters β, γ;

Initialization: $t = 0$, $\mathbf{X}^0 = (\mathbf{D}^T\mathbf{D})^{-1}\mathbf{D}^T\mathbf{Y}$;
for $t = 1, 2, \cdots, T$ **do**
  Divide $\mathbf{X}^t$ into a set of overlapping patches with size $\sqrt{b} \times \sqrt{b}$;
  for Each patch $\mathcal{X}_i$ in $\mathbf{X}^t$ **do**
    Find nonlocal similar patches group of the representation coefficients $\mathcal{R}_i\mathbf{X}$ via KNN algorithm;
    SVD for $\mathcal{R}_i\mathbf{X}$;
    Update $\mathbf{L}_i^*$ by computing (12);
  end for
  ADMM:
  Initialization: $\mathbf{Z}_1^0 = 0$, $\mathbf{Z}_2^0 = 0$, $\mathbf{V}_1^0 = 0$, $\mathbf{V}_2^0 = 0$, $\mathbf{V}_3^0 = 0$, $\sigma = 1$, $\tau = 1.618$.
  Update $\mathbf{X}^*$ by computing (17);
  Update $\mathbf{E}^*$ by computing (19);
  Update $\mathbf{Z}_1^*$ by computing (22);
  Update $\mathbf{Z}_2^*$ by computing (24);
  Update the Lagrangian dual variables $\mathbf{V}_1$, $\mathbf{V}_2$, and $\mathbf{V}_3$ by computing (25);
end for
Obtain the optimal solution $\mathbf{E}^*$;
Compute the anomaly detection map $\mathbf{T}$ by (26);

Output: Anomaly detection map **T.**

specifically, the anomaly map $\mathbf{T}$ is determined by the response of $\mathbf{E}$, as follows:

$$\mathbf{T}(\mathbf{x}_i) = \|\mathbf{E}(:, i)\|_2 > \epsilon. \tag{26}$$

The ith pixel is classified as an anomaly pixel if (26) exceeds a predefined threshold ϵ. The entire procedure of the proposed framework is outlined in Algorithm 1.

C. Computational Complexity Analysis

The computational complexity for updating $\mathbf{L}_i^*$ in each iteration is $O(nb^3)$. The complexity for updating $\mathbf{X}^*$ is $O(m^2L) + O(m^3) + O(nmL) = O(nm \cdot \max(L, m))(n \gg L, m)$. For computing $\mathbf{E}^*$ and $\mathbf{Z}_1^*$, the complexities are $O(nmL)$ and $O(m^3)$. The complexity for updating $\mathbf{Z}_2^*$ is $O(dmn)$ with d being the depth of the FFDNet. For updating $\mathbf{V}_1$, $\mathbf{V}_2$, and $\mathbf{V}_3$, the computational complexities are $O(mn)$, $O(mn)$, and $O(nmL)$, respectively. Therefore, the overall computational complexity for Algorithm 1 is $O(n(b^3 + m \cdot \max(L, m)))$.

IV. Numerical Experiments

In this section, we present the experiments performed on several datasets to evaluate the proposed method. The results of the proposed method have been compared with those obtained with six model-drive-based methods including RX [8], RPCA-RX [29], TPCA [36], LSMAD [30], LRASR [31], and GTVLRR [35], as well as four data-drive-based methods including RGAE [16], BockNet [20], MPDA [21], and MFT-Net [23]. For the proposed method, the size of nonlocal cubic $\sqrt{b}$ and the number of nonlocal similar cubics k were set to 7 and 300, respectively [72]. In the background dictionary construction step, the number of clusters M and selected P atoms were set to 15 and 20, respectively [31]. For a fair comparison, the regularization parameters of all the compared approaches were set as the recommended values or tuned to obtain the best detection performance.

A. Evaluation Metrics

The detection performance of the proposed method was evaluated both qualitatively and quantitatively in comparison to ten state-of-the-art methods. For the qualitative assessment, we visually compared the detection maps produced by each method. Quantitative evaluation was conducted using the receiver operating characteristic (ROC) curve, the area under the ROC curve (AUC), and the separability boxplot as objective metrics to measure detection accuracy. The ROC curve illustrates the relationship between the detection rate (DR) and false alarm rate (FAR), with the horizontal axis representing FAR and the vertical axis representing DR. The DR refers to the percentage of correctly detected target pixels N_D that occupy the number of true anomalous pixels N_A, while the FAR refers to the percentage of false alarm pixels N_F that occupy the total number of pixels N_T. Specifically, DR and FAR can be represented as

$$\text{DR} = \frac{N_D}{N_A}, \quad \text{FAR} = \frac{N_F}{N_T}. \tag{27}$$

In general, the closer the ROC curve is to the upper leftmost corner indicating a higher DR and lower FAR, the better the detection performance. The AUC value is calculated as the area under the ROC curve along the horizontal axis of the coordinate system. An AUC value closer to 1 indicates better performance of the detection method. The separability boxplot displays the statistical range of abnormal response values at pixel locations within the background and anomaly categories, represented by two separate boxes—blue for the background and red for the anomaly—each spanning a range from 10% to 90%. Ideally, if the detector effectively suppresses the background and extrudes the anomalies, the blue box will appear low on the plot, narrow in width, and distinctly separated from the red box, indicating strong separability between the two categories.

B. Data Description

1) San Diego Dataset: The first hyperspectral dataset used in the experiment is the San Diego dataset,[1] captured by the airborne visible/infrared imaging spectrometer (ARIVIS) sensor over San Diego, USA. This dataset consists of 400×400 pixels with 224 spectral bands, uniformly distributed from 0.4 to 2.5 μm. Two subimages, each with 100 × 100 pixels, were selected and named San Diego-I and San Diego-II. After removing water vapor absorption and noisy bands, 189 spectral bands remained for analysis. In both the scenarios, three airplanes are treated as anomalies, while roofs, buildings, grass, and shadows comprise the background.

[1] http://xudongkang.weebly.com/data-sets.html

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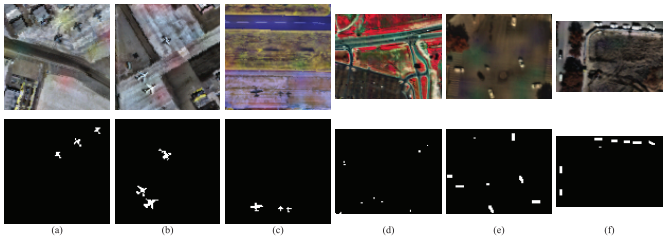


Fig. 2. Pseudocolor images and ground-truth maps of the four considered HSI datasets. (a) San Diego-I. (b) San Diego-II. (c) Gulfport. (d) Hydice-Urban. (e) AIR-HI-Vehicle-I. (f) AIR-HI-Vehicle-II.

2) Gulfport Dataset: The second hyperspectral dataset we used is the Gulfport dataset[1], which was captured by the ARIVIS sensor over Gulfport, USA. These data consist of 100×100 pixels, measured in 224 bands ranging from 0.4 to 2.5 μm. After removing bands affected by dense water vapor absorption and atmospheric effects, 191 spectral bands remained for analysis. In this scene, anomalies are flights of different heights, while buildings and roads form the background.

3) Hydice-Urban Dataset: The third hyperspectral dataset used for detection is the Hydice-Urban dataset,[2] which was captured by the hyperspectral digital imagery collection experiment (HYDICE) sensor over California, USA. These data contain 80×100 pixles with 175 bands, covering wavelengths from 0.4 to 2.5 μm. The scene includes a construction area, a vegetation area, and several roads with a few vehicles. The anomalies in this dataset are man-made objects of varying sizes, such as roofs and vehicles.

4) AIR-HI Dataset: The last aerial dataset used for detection is the AIR-HI dataset,[3] which was captured by the visible and near-infrared imaging spectrometer developed by the Shanghai Institute of Technical Physics, Chinese Academy of Sciences, over Xiong'An New Area, China. This dataset was acquired at a flight altitude of 2000 m, with a spatial resolution of 0.525 m per pixel. It contains 3751×2746 pixels across 253 spectral bands, uniformly distributed from 0.4 to 1.0 μm. Two subimages, with dimensions 80×100 and 80×120 pixles, were chosen and named AIR-HI-Vehicle-I and AIR-HI-Vehicle-II. In both the cases, several vehicles are regarded as anomalies, while grass, shadows, and roads make up the background.

The pseudocolor images and ground-truth maps of all the datasets are presented in Fig. 2.

C. Results and Analysis

1) Detection Maps: Figs. 3–8 display the HAD maps obtained by different methods across all datasets. For the San Diego-I dataset, it can be observed that the RX and RPCA-RX methods failed to detect the airplane, while the TPCA method struggled to accurately capture the airplane's contours. Although the LSMAD, RGAE, BockNet, and MFTNet methods provide a clear background, they do not effectively highlight the airplane. The MPDA method highlights both the anomalies and a portion of the background, while losing the contour of the anomalous target. Among the dictionary-based approaches, both the LRASR and GTVLRR methods successfully detect the anomalies, with the proposed NLDPAD method delivering the brightest and most distinct airplane silhouette. In the San Diego-II dataset, all methods effectively detect the anomalies. However, the proposed NLDPAD method offers a clearer and more defined shape of the airplane compared with the other methods. For the Gulfport dataset, the RX and RPCA-RX methods fail to detect the two small airplanes. Although the TPCA, LSMAD, LRASR, RGAE, BockNet, and MFTNet methods identified all the anomalies, but again failed to capture the contours of the two small airplanes. Compared with the GTVLRR and MPDA methods, the detection map offered by the proposed NLDPAD method provides better background suppression. In the Hydice-Urban dataset, the RX, RPCA-RX, LSMAD, RGAE, and MFTNet methods produced clean detection maps but missed a few targets. The FAR of the detection images provided by the TPCA and MPDA methods is too high. The LRASA, GTVLRR, BockNet, and NLDPAD methods generated similar detection maps, with the NLDPAD method demonstrating the best overall performance. For the AIR-HI-Vehicle-I and AIR-HI-Vehicle-I datasets, the proposed NLDPAD method obtains much clearer detection maps than the other methods, highlighting its effectiveness across diverse datasets.

2) ROC Curves: The ROC curves obtained by different methods across all datasets are shown in Fig. 9. Considering the San Diego-I dataset as an example, the proposed NLDPAD method achieves the highest probability of detection under a low probability of false alarm and achieves 100% detection when the probability of false alarm exceeds 5×10^{-2}. Another example is the Hydice-Urban dataset; although the detection probability of the proposed method is lower than the BockNet method when the probability of false alarm is below 10^{-3}, it outperforms all other methods in the remaining range. Overall, these results highlight the superiority of the proposed NLDPAD method, demonstrating its effectiveness over both the model-driven and data-driven methods across different detection scenarios.

3) AUC Values: Table I lists the AUC values of different methods for all datasets, and the best AUC scores are highlighted in bold. As expected, the proposed NLDPAD method achieves the highest AUC scores for the San Diego, Gulfport, and Hydice-Urban datasets, outperforming other compared methods. Notably, for the AIR-HI-Vehicle-I dataset, the BockNet method yields a slightly higher AUC than the NLDPAD method, while for the AIR-HI-Vehicle-II dataset, the HADGSM method surpasses BockNet by an AUC margin of 0.03. The computation costs of all the detection methods are also compared in Table I. The statistic-based methods RX and RPCA-RX are relatively fast due to their straightforward implementation, and the TPCA and LSMAD methods have certain advantages over the rest of the methods. The LRR-based methods, LRASR and GTVLRR, require more time as they involve hundreds of iterations to reach convergence. The deep-learning-based methods spend more time in the training

[2]https://github.com/mahdiyousofun/AED-algorithm/tree/master/Data
[3]https://github.com/MinghuaWang123/DSR-ADNR

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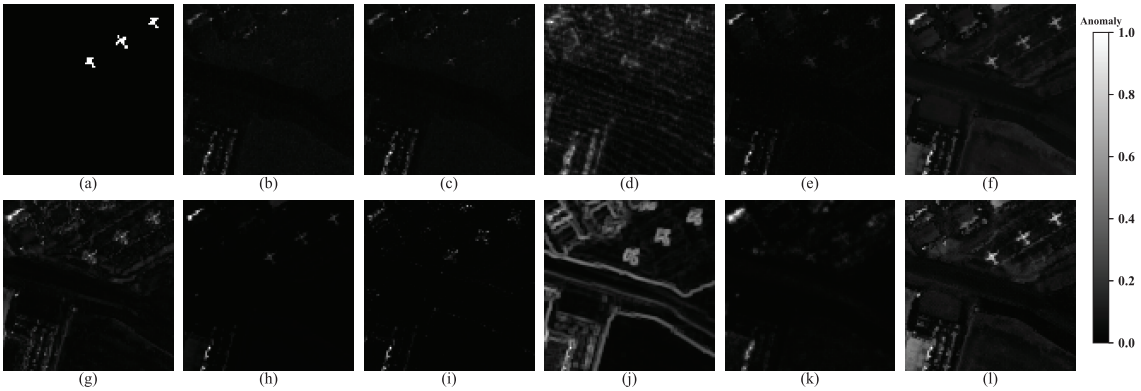


Fig. 3. HAD colored maps obtained by different methods for the San Diego-I dataset. (a) Ground truth (AUC value). (b) RX (0.8886). (c) RPCA-RX (0.9165). (d) TPCA (0.8840). (e) LSMAD (0.9780). (f) LRASR (0.9916). (g) GTVLRR (0.9880). (h) RGAE (0.9903). (i) BockNet (0.9796). (j) MPDA (0.9724). (k) MFTNet (0.9677). (l) NLDPAD (0.9927).

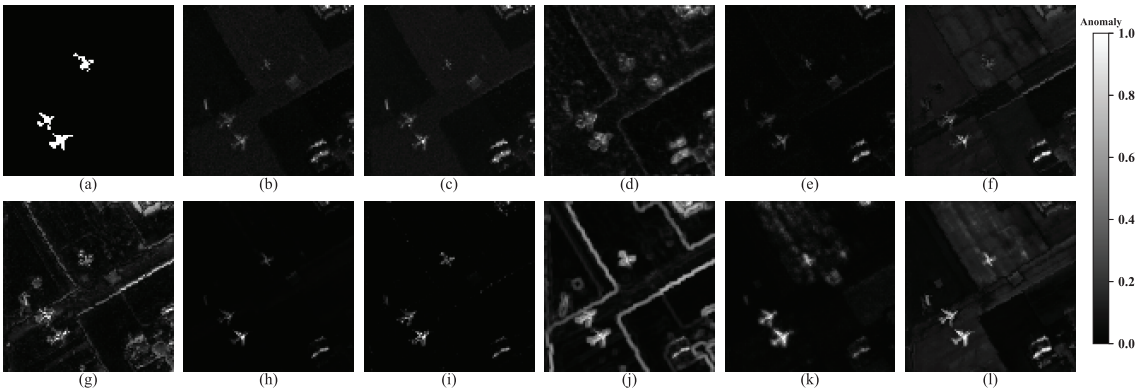


Fig. 4. HAD colored maps obtained by different methods for the San Diego-II dataset. (a) Ground truth (AUC value). (b) RX (0.9403). (c) RPCA-RX (0.9294). (d) TPCA (0.9422). (e) LSMAD (0.9688). (f) LRASR (0.9125). (g) GTVLRR (0.9373). (h) RGAE (0.9902). (i) BockNet (0.9878). (j) MPDA (0.9603). (k) MFTNet (0.9875). (l) NLDPAD (0.9933).

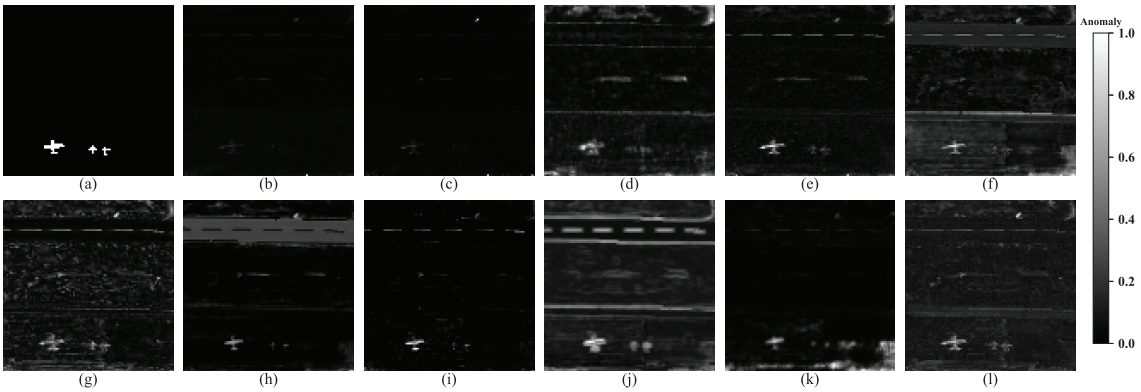


Fig. 5. HAD colored maps obtained by different methods for the Gulfport dataset. (a) Ground truth (AUC value). (b) RX (0.9526). (c) RPCA-RX (0.9627). (d) TPCA (0.9432). (e) LSMAD (0.9864). (f) LRASR (0.9425). (g) GTVLRR (0.9832). (h) RGAE (0.9126). (i) BockNet (0.9883). (j) MPDA (0.9800). (k) MFTNet (0.9468). (l) NLDPAD (0.9919).

phase, with RGAE being more time-consuming. The proposed NLDPAD method also has a relatively high runtime, ranking second highest due to the time-consuming process of nonlocal similar patch searches.

4) Separability Boxplots: Separability boxplots are used to facilitate a more in-depth assessment of the separability between the anomalies and the background for all the detectors. The separability maps are shown in Fig. 10, where blue is for the distributions of background and red is for the distributions of anomalies. From Fig. 10(a) to (d), one can clearly observe that the proposed NLDPAD method can ensure a more pronounced gap between the two boxes compared with other detectors, indicating a superior separation between anomalies and background. In Fig. 10(e)–(f), although the

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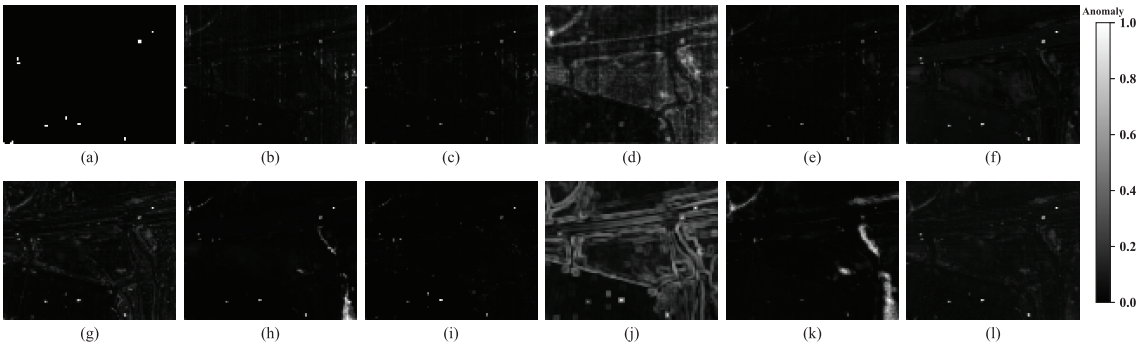


Fig. 6. HAD colored maps obtained by different methods for the Hydice-Urban dataset. (a) Ground truth (AUC value). (b) RX (0.9857). (c) RPCA-RX (0.9843). (d) TPCA (0.8242). (e) LSMAD (0.9904). (f) LRASR (0.9765). (g) GTVLRR (0.9906). (h) RGAE (0.9552). (i) BockNet (0.9863). (j) MPDA (0.9013). (k) MFTNet (0.8734). (l) NLDPAD (0.9919).

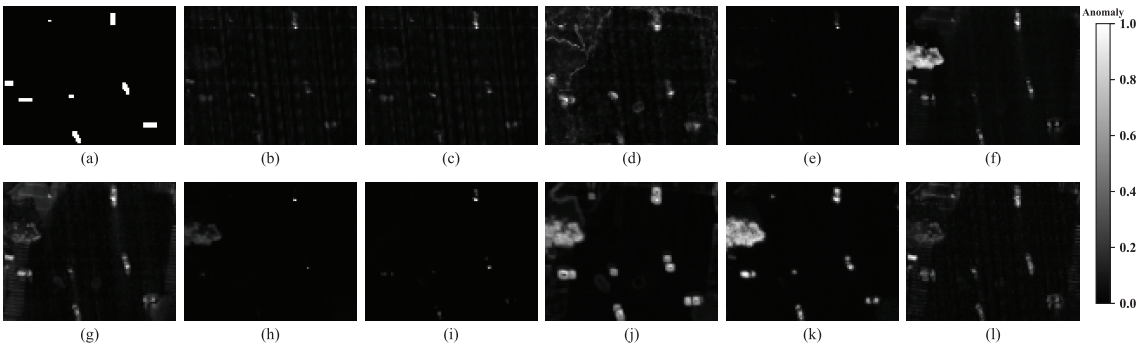


Fig. 7. HAD colored maps obtained by different methods for the AIR-HI-Vehicle-I dataset. (a) Ground truth (AUC value). (b) RX (0.9586). (c) RPCA-RX (0.9538). (d) TPCA (0.9653). (e) LSMAD (0.9659). (f) LRASR (0.8988). (g) GTVLRR (0.9622). (h) RGAE (0.9305). (i) BockNet (0.9724). (j) MPDA (0.9644). (k) MFTNet (0.9605). (l) NLDPAD (0.9673).

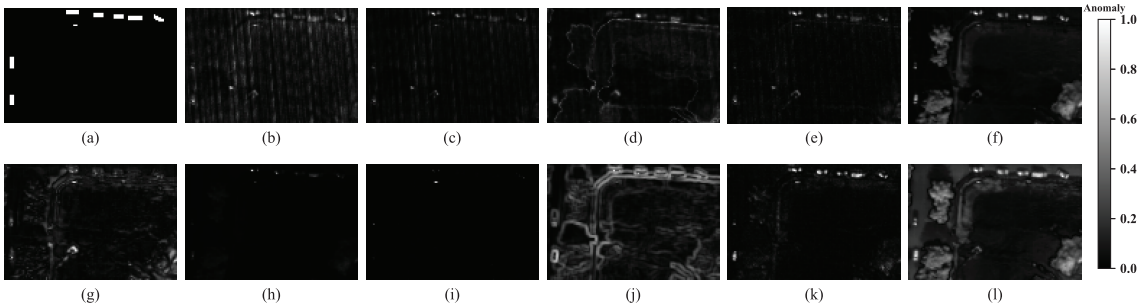


Fig. 8. HAD colored maps obtained by different methods for the AIR-HI-Vehicle-II dataset. (a) Ground truth (AUC value). (b) RX (0.9065). (c) RPCA-RX (0.9052). (d) TPCA (0.8279). (e) LSMAD (0.9323). (f) LRASR (0.8694). (g) GTVLRR (0.9014). (h) RGAE (0.9318). (i) BockNet (0.9031). (j) MPDA (0.9261). (k) MFTNet (0.9292). (l) NLDPAD (0.9337).

gap between the two boxes has vanished for all detectors, the red anomaly boxes of the proposed NLDPAD method are positioned higher, indicating a stronger distinction for anomalies. These observations align with the visual analysis results discussed earlier and further demonstrate the superiority of the proposed method for the detection task.

D. Parameters' Sensitivity Analysis

The proposed NLDPAD method involves two regularization parameters β and γ. The parameter γ balances the contributions of the background and anomalies, playing a critical role in anomaly detection. Fig. 11 shows the AUC values as a function of the parameter γ on different datasets. For the San Diego and AIR-HI datasets, the NLDPAD method achieves higher and relatively stable AUC values when γ is below 10^{-2}; however, the AUC declines as γ increases beyond this range. For the Gulfport and Hydice-Urban datasets, the AUC values increase as γ increases and reach a peak when γ is close to 1. Based on these observations, we set $\gamma = 10^{-3}$ for the San Diego and AIR-HI datasets, $\gamma = 1$ for the Gulfport dataset, and $\gamma = 0.8$ for the Hydice-Urban dataset. The parameter β is related to the input's standard deviation of Gaussian noise, and we empirically set it to 1.

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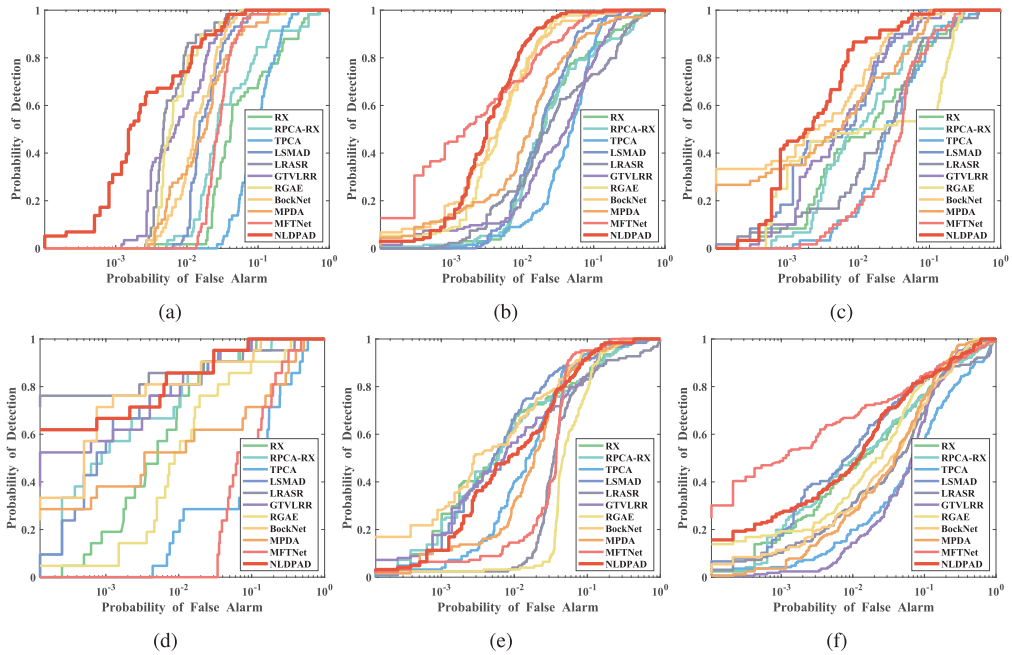


Fig. 9. ROC curves obtained by different methods for different datasets. (a) San Diego-I. (b) San Diego-II. (c) Gulfport. (d) Hydice-Urban. (e) AIR-HI-Vehicle-I. (f) AIR-HI-Vehicle-II.

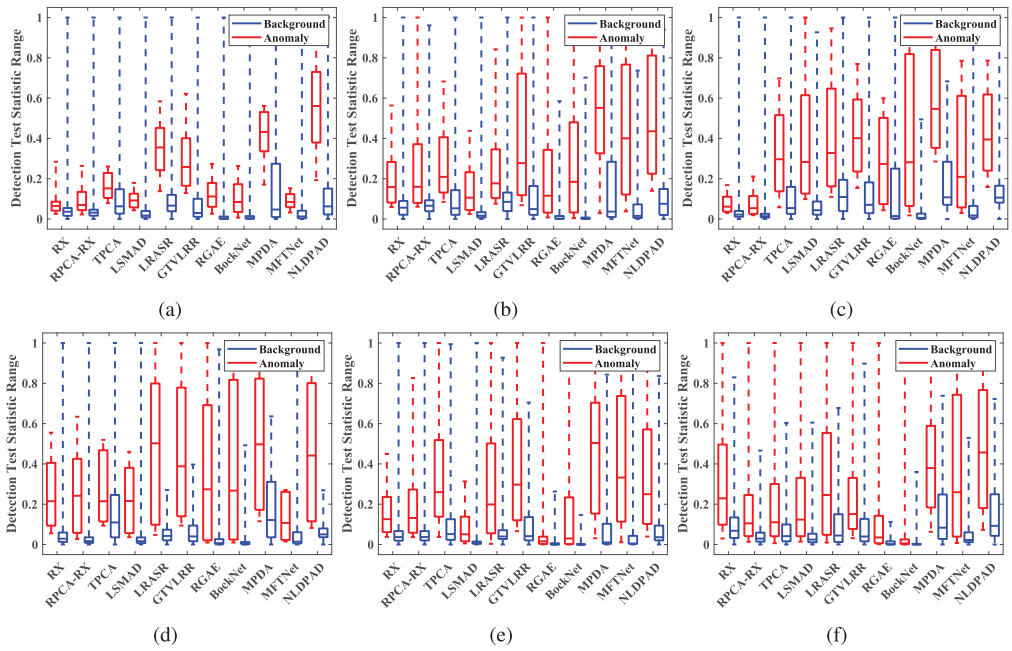


Fig. 10. Separability maps obtained by different methods on different datasets. (a) San Diego-I. (b) San Diego-II. (c) Gulfport. (d) Hydice-Urban. (e) AIR-HI-Vehicle-I. (f) AIR-HI-Vehicle-II.

E. Ablation Study

The impact of individual components within the NLDPAD model is evaluated by analyzing their effect on detection performance across all datasets. To verify, we use the low-rank and deep priors for anomaly detection (LRDPAD) model, which applies the low-rank and deep regularizers directly to the representation coefficients $\mathbf{X}$, instead of the nonlocal similar patch group of the representation coefficients $\mathcal{R}_i\mathbf{X}$ as done in NLDPAD. Furthermore, we consider two variants: the NLAD without weighting (denoted as NLAD$_W$) and NLAD models, both of which use the same paradigm as NLDPAD except for removing the deep prior term. Table II presents the AUC values and running times for NLAD$_W$, NLAD, LRDPAD, and NLDPAD methods on different datasets. As shown in Table II, the proposed NLDPAD method consistently achieves higher AUC values than the other methods across all datasets, demonstrating that the integration of nonlocal self-similarity and deep priors leads to significantly improved detection accuracy compared with using a single prior. In terms of computational efficiency, LRDPAD is the fastest

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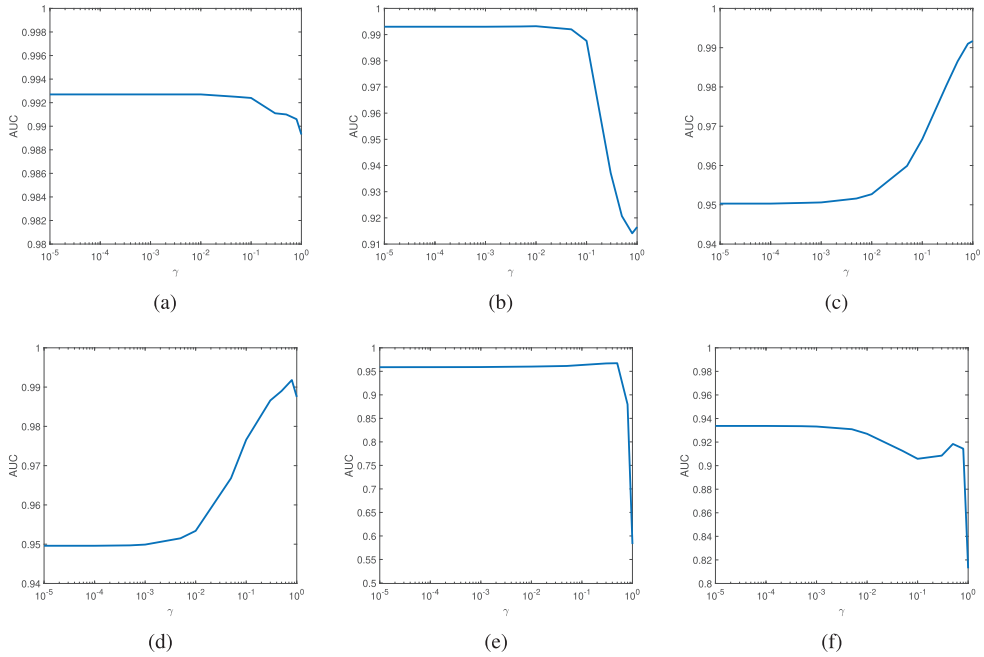


Fig. 11. Parameter sensitivity analysis for the regularization parameter γ on different datasets. (a) San Diego-I. (b) San Diego-II. (c) Gulfport. (d) Hydice-Urban. (e) AIR-HI-Vehicle-I. (f) AIR-HI-Vehicle-II.

TABLE I
AUC VALUES AND RUNNING TIME OF DIFFERENT METHODS FOR DIFFERENT DATASETS

Data	Metric	Method										
		RX	RPCA-RX	TPCA	LSMAD	LRASR	GTVLRR	RGAE	BockNet	MPDA	MFTNet	NLDPAD
San Diego-I	AUC	0.8886	0.9165	0.8840	0.9780	0.9916	0.9880	0.9903	0.9796	0.9724	0.9677	**0.9927**
	Time (Sec.)	**0.25**	3.09	12.87	9.27	27.99	118.90	354.16	49.79	99.05	199.45	209.11
San Diego-II	AUC	0.9403	0.9294	0.9422	0.9688	0.9125	0.9373	0.9902	0.9878	0.9603	0.9875	**0.9933**
	Time (Sec.)	**0.24**	3.05	12.74	8.40	32.51	128.90	340.56	50.49	101.36	333.19	340.33
Gulfport	AUC	0.9526	0.9627	0.9432	0.9864	0.9425	0.9832	0.9126	0.9883	0.9800	0.9468	**0.9919**
	Time (Sec.)	**0.05**	2.97	13.73	8.15	31.17	157.20	355.42	50.03	97.00	327.79	150.93
Hydice-Urban	AUC	0.9857	0.9843	0.8242	0.9904	0.9765	0.9906	0.9552	0.9863	0.9013	0.8734	**0.9919**
	Time (Sec.)	**0.04**	2.15	9.26	5.60	25.84	110.90	279.88	49.06	77.84	259.80	131.72
AIR-HI-Vehicle-I	AUC	0.9586	0.9538	0.9653	0.9659	0.8988	0.9622	0.9305	**0.9724**	0.9644	0.9605	0.9673
	Time (Sec.)	**0.04**	6.61	13.04	8.96	19.98	76.80	293.33	18.34	188.14	295.26	245.33
AIR-HI-Vehicle-II	AUC	0.9065	0.9052	0.8279	0.9323	0.8694	0.9014	0.9318	0.9031	0.9261	0.9292	**0.9337**
	Time (Sec.)	**0.09**	9.90	19.77	11.08	27.83	313.70	336.44	78.37	105.74	139.10	197.75

among the compared methods, while NLAD_W and NLAD require less than half the runtime of NLDPAD. This highlights a clear tradeoff between detection performance and computational cost. The increased runtime of NLDPAD is primarily attributed to the nonlocal prior, which involves time-consuming searches for similar patch groups. As expected, incorporating more prior information not only improves accuracy but also increases computational demand.

F. Initial Values' Analysis

To analyze the impact of initialization, we compared the AUC values obtained by the proposed NLDPAD method using two different initialization strategies on the San Diego-I dataset. In Case I, the initial values of $\mathbf{X}^0$, $\mathbf{Z}_1^0$, $\mathbf{Z}_2^0$, $\mathbf{V}_1^0$, $\mathbf{V}_2^0$, and $\mathbf{V}_3^0$ are all set equal to zero, resulting in an AUC of 0.9828. In Case II, the initial values are set according to the strategy described in Algorithm 1, yielding an improved

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TABLE II
AUC VALUES AND RUNNING TIME OF $NLAD_W$, NLAD, AND NLDPAD FOR DIFFERENT DATASETS

Data	Metric	Method			
		$NLAD_W$	NLAD	LRDPAD	NLDPAD
San Diego-I	AUC	0.9819	0.9903	0.9802	**0.9927**
	Time (Sec.)	105.73	106.31	**78.69**	209.11
San Diego-II	AUC	0.9610	0.9698	0.9591	**0.9933**
	Time (Sec.)	112.33	113.74	**87.63**	340.33
Gulfport	AUC	0.9738	0.9808	0.9788	**0.9919**
	Time (Sec.)	75.55	75.63	**58.55**	150.93
Hydice-Urban	AUC	0.9727	0.9820	0.9755	**0.9919**
	Time (Sec.)	60.23	61.45	**49.21**	131.72
AIR-HI-Vehicle-I	AUC	0.9511	0.9523	0.9466	**0.9673**
	Time (Sec.)	177.78	178.65	**109.15**	245.33
AIR-HI-Vehicle-II	AUC	0.9005	0.9115	0.8849	**0.9336**
	Time (Sec.)	97.95	98.45	**67.66**	197.75

AUC of 0.9927. These results indicate that an appropriate initialization strategy can positively influence the detection performance of the proposed method.

V. CONCLUSION

In this article, we introduced a novel framework, NLDPAD, to enhance the accuracy of HAD. The proposed method uses nonlocal low-rank modeling, effectively capturing NSS and GCS within the subspace of HSIs. By incorporating a deep denoiser prior, the framework leverages external information to further improve detection performance. The results highlight the potential of the NLDPAD framework for advancing anomaly detection in hyperspectral imaging. Future work will focus on the exploration of different types of nonlocal or deep priors, such as the two-stream CNN [73], to further capture various types of information with HSIs. Reducing the computational cost of nonlocal modeling could also be a promising avenue for future research.

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